

Program9

Organize kubernetes pods using labels and namespace

Project Structure:

```
-----program9
      ----- namespace.yaml
      ----- pod.yaml
```

Step1: create a folder

Step2: create a namespace.yaml and pod.yaml

>> nano namespace.yaml

>> nano pod.yaml

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx: ~/program9  ×  rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ mkdir program9
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd program9
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ nano pod.yaml
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ nano namespace.yaml
```

Step3: Apply namespace and pod.yaml file



```
GNU nano 2.2.2 namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step4: verify pods by namespace

```
GNU nano 7.2 pods.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
```

Help Write Out Where Is Cut Execute Location Undo Set Mark
Exit Read File Replace Paste Justify Go To Line Redo Copy

Step5: Apply namespace and pod.yaml file

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl apply -f namespace.yaml
namespace/prod created
namespace/dev created
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl apply -f pod.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step6: Verify the pod by namespace

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           46s
frontend-pod  1/1     Running   0           46s
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           50s
frontend-pod  1/1     Running   0           50s
```

Step7: Check the status

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
hello-cron-29425706-qtmj2	0/1	Completed	0	6d21h
hello-cron-29435647-phgc6	0/1	Completed	0	80s
hello-cron-29435648-2fsjr	0/1	Completed	0	27s

Step8: Filter pod by labels

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl get pods -n dev -l app=frontend
```

NAME	READY	STATUS	RESTARTS	AGE
frontend-pod	1/1	Running	0	2m13s

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program9$ kubectl get pods -n dev -l app=backend
```

NAME	READY	STATUS	RESTARTS	AGE
backend-pod	1/1	Running	0	2m20s

Step9: Clear all the process and delete all pods and services.